Session 7 homework — the scoreboard, and what a pretrained CNN adds

Computer Vision & Speech Recognition · Joshua Cole · MILK10k skin-lesion classification (Benign / Indeterminate / Malignant) · seed 42 splits

Repository: https://github.com/jcole-jpg/Computer-Vision-and-Speech-Recognition

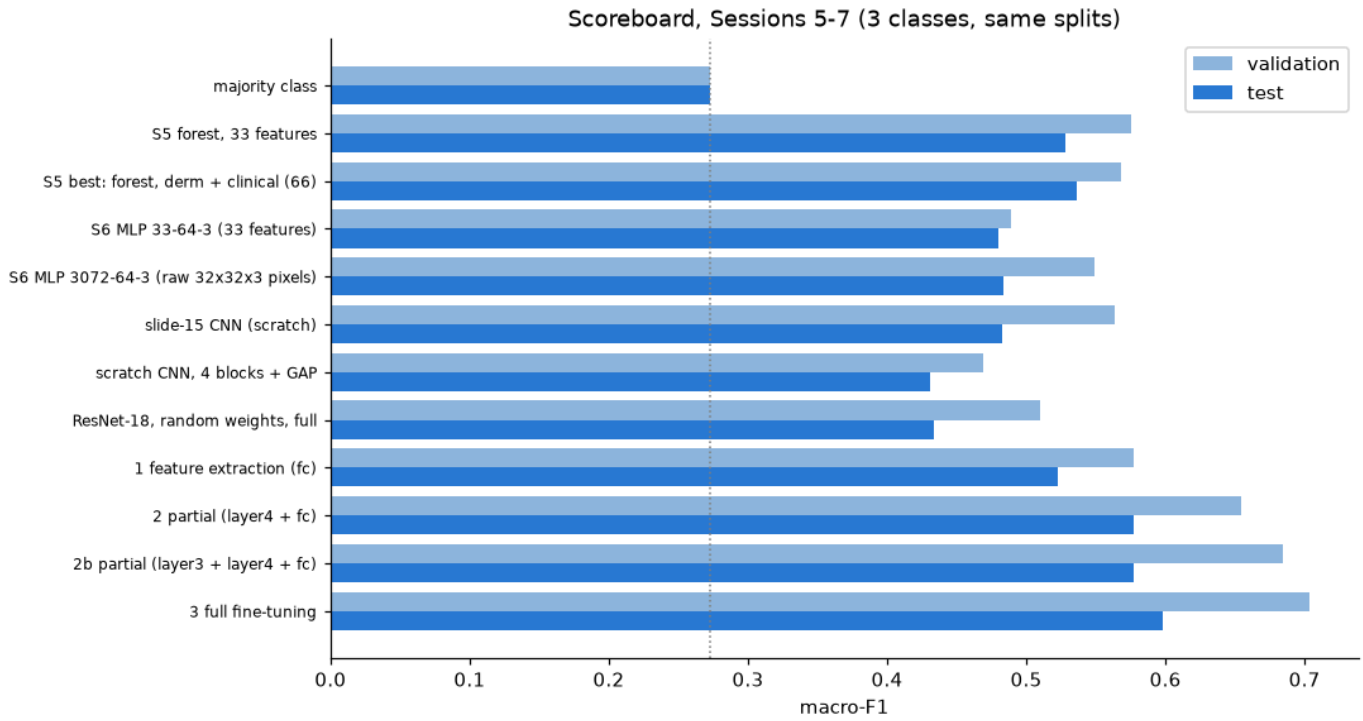
Code that completes the homework (slide 30): `notebooks/07_cnn.ipynb` (the experiment, executed, every table and figure; PDF: `notebooks/07_cnn.pdf`) · `src/milk10k/cnn.py` (tensors, augmentation, networks, fine-tuning strategies, cached training runs) · `scripts/build_session7_notebook.py` (generates and runs the notebook) · `tests/test_cnn.py` (13 tests). Raw outputs: `artifacts/session07/scoreboard.csv`, `artifacts/session07/mini_challenge.csv`, `artifacts/session07/scores.json`, `artifacts/session07/runs/`. Earlier scoreboard rows: `notebooks/05_classical_baseline.ipynb`, `notebooks/06_neural_networks.ipynb`.

Result

The model to bring to Session 8 is ResNet-18 pretrained on ImageNet and fine-tuned in full (strategy 3, the optional homework): validation macro-F1 0.703, test macro-F1 0.598 (95 % bootstrap interval [0.530, 0.662]), test accuracy 0.754. It was chosen by validation macro-F1 alone, the highest of everything in Sessions 5–7, and the test split was scored once, after every choice was fixed. Always predicting the majority class scores 0.273.

The scoreboard, Sessions 5–7 (same lesion-level splits; macro-F1)

session	model	trainable params	validation	test	test accuracy	Malignant recall
5	majority class	0	0.273	0.273	0.694	–
5	Session 5 forest, 33 features	–	0.575	0.528	–	–
5	Session 5 best: forest, derm + clinical (66)	–	0.568	0.536	0.744	–
6	Session 6 MLP 33-64-3 (33 features)	2,371	0.489	0.480	0.637	–
6	Session 6 MLP 3072-64-3 (raw 32x32x3 pixels)	196,867	0.549	0.484	0.724	–
7	slide-15 CNN (scratch)	306,147	0.564	0.483	0.717	0.779
7	scratch CNN, 4 blocks + GAP	389,667	0.469	0.431	0.600	0.621
7	ResNet-18, random weights, full	11,178,051	0.510	0.434	0.662	0.777
7	1 feature extraction (fc)	1,539	0.577	0.523	0.700	0.742
7	2 partial (layer4 + fc)	8,395,267	0.654	0.577	0.768	0.842
7	2b partial (layer3 + layer4 + fc)	10,494,979	0.684	0.577	0.737	0.739
7	3 full fine-tuning **[PICK]**	11,178,051	0.703	0.598	0.754	0.788



How to read it

- **Pretraining is what matters.** The same ResNet-18 from random weights scores 0.510 on validation and 0.434 on test; from ImageNet weights, fully fine-tuned, 0.703 and 0.598. Small networks trained from scratch on 3,742 lesions (slide-15 net 0.483,

four-block net 0.431 on test) do no better than Session 6's MLPs (0.480, 0.484).

- **More unfrozen layers, better score, higher cost:** `fc` only 0.523 (8.7 s/epoch), `layer4 + fc` 0.577 (11.0 s), `layer3 + layer4 + fc` 0.577 (15.0 s), everything 0.598 (31.0 s) on test. The gaps between the last three are inside the seed noise measured in the mini-challenge (std 0.023 on validation), so `layer4 + fc` is statistically indistinguishable from the pick at a third of the training time.
- **Against Session 5's forest** (test 0.536, interval [0.472, 0.601]) the pick is +0.062; the intervals overlap, so this is a likely improvement on 749 test lesions, not a proven one.
- **Validation is optimistic** for every model (0.703 against 0.598 for the pick): the best epoch and the best model were both chosen on it. The test number is the one to quote.
- **What it gets wrong:** per-class recall on test is Benign 0.70, Indeterminate 0.41, Malignant 0.79; 96 of 520 malignant lesions are called benign. At argmax the cost per lesion (missed malignant 50, unnecessary work-up 1) is 6.65, well above the 0.40 of always answering "Malignant"; the matrix calls for a screening threshold below 0.5, which is not tuned here.

EfficientNet or DenseNet versus ResNet

DenseNet-121 differs from ResNet-18 in how it reuses earlier features: where a ResNet block adds its input to its output (`x + f(x)`), each DenseNet layer receives the concatenated feature maps of all earlier layers in its block, so features are reused instead of relearned and the network gets by with 8.0 M parameters against 11.7 M, at the price of more memory per layer. EfficientNet-B0 keeps ResNet's skip connections but replaces its plain 3×3 blocks with MBConv blocks (depthwise-separable convolutions with squeeze-and-excitation) and sets depth, width and input resolution together by one scaling rule, reaching 5.3 M parameters. Parameter counts are from torchvision (notebook section 7); neither network was trained here, so no accuracy claim is made for them.

Protocol and assumptions

- Dermoscopic image of each lesion only, stretched to 224×224, ImageNet normalisation, flips / quarter turns / mild brightness and contrast on training batches only; class-weighted cross-entropy; Adam; **10 epochs** for every network; the best-validation epoch is kept.
- Pretrained layers train at 1e-4, the new head (and every from-scratch network) at 1e-3; frozen layers keep their BatchNorm statistics.
- Seed 0 for the scoreboard; seeds 1 and 2 only for the mini-challenge, so strategy 3's number comes from a single run. Apple MPS training is not bit-for-bit repeatable; the notebook caches every run, so re-executing it reproduces the numbers above.
- Sessions 5 and 6 rows are read from `artifacts/session05` and `artifacts/session06` (same splits), not retrained.